

UCCC 2063 Algorithm Analysis

Quiz

Time allowed: 1 hour

Full mark = 75

Question One: [Analysis of Algorithm]

Answer the following questions in terms of n using the big-Oh notation. No explanation is needed.

- a) Consider the algorithm below.

```
counter = 0;
for i = 1 to  $n^4$ 
    for j = 1 to  $n^2$ 
        for k = 1 to  $n$ 
            counter = counter + 1;
```

- (i) What is the value of counter at the end of the algorithm?
- (ii) What is the time complexity of the algorithm?

(4 marks)

- b) Consider the algorithm below.

```
counter = 0;
for i = 1 to  $n^4$ 
    for j = 1 to  $n^2$ 
        if  $i$  is equal to  $j$ 
            for k = 1 to  $n$ 
                counter = counter + 1;
```

- (i) How many times is the if-condition tested ?
- (ii) How many times is the if-condition tested TRUE ?
- (iii) What is the value of counter at the end of the algorithm?
- (iv) What is the time complexity of the algorithm?

(8 marks)

- c) Consider the algorithm below.

```
counter = 0;
for i = 1 to  $n^4$ 
    for j = 1 to  $n^2$ 
        if  $j$  is equal to  $\log i$ 
            for k = 1 to  $n$ 
                counter = counter + 1;
```

- (i) How many times is the if-condition tested ?
- (ii) How many times is the if-condition tested TRUE ?
- (iii) What is the value of counter at the end of the algorithm?
- (iv) What is the time complexity of the algorithm?

(8 marks)

Question Two: [Asymptotic Notations]

You are given the definitions of Big-Oh and Big- Ω as below.

$$O(g(n)) = \{f(n) \mid \exists c, n_0 \bullet \forall n > n_0 \bullet f(n) < c g(n)\}.$$

$$\Omega(g(n)) = \{f(n) \mid \exists c, n_0 \bullet \forall n > n_0 \bullet f(n) > c g(n)\}.$$

- a) Using the definition of Big-Oh, prove $4n^3 + 3\sqrt{n} = O(n^3)$. (10 marks)
- b) Using the definition of Big- Ω , prove $4n^3 + 3\sqrt{n} \neq \Omega(n^{3.1})$. (15 marks)
- c) Using the definition of Big-Oh and Big- Ω , prove $O(n^2) \cap \Omega(n^5) = \emptyset$. (15 marks)
- d) Without using the stirling's formula, prove $\log n! = \Omega(n \log n)$. (15 marks)

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